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CropStetho

An Artificial Intelligence Based Crop Diagnosis tool which take image of Leaf of crop as input and Predict the Disease with the Information on that disease and solution.

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Inspiration

The challenge topic: Diseases and Pests

How can technology reduce the economic impact of plant diseases and pests by improving farmers' ability to quickly and accurately diagnose and respond to plants/trees/vines with disease or pest issues?

The following are the problems faced by farmers :

1. Agriculture plays a critical role in providing food supply for growing population of the world.
2. Annual global food supply loss due to plant disease is 40%, on average.
3. In most of the countries, smallholder farmers generate more than 80% of the agricultural production. For them, the loss of crops has devastating consequences.
4. Sometimes, farmers can lose almost 100% of their crop due to plant disease. This makes crop disease a major threat to food security around the world. after looking at these problem comes the idea of CropStetho - An Artificial Intelligence Based Crop Diagnosis tool which take image of Leaf of crop as input and Predict the Disease with the Information on that disease and solution.

Proposed Solution

Timely and correct identification of a disease when it first appears is a critical step for efficient disease management.

An app that the farmer can have their smartphone with the following capabilities: *Identify the type of plant *Identify if the plant has the disease *If the leaves are affected by a disease, classify the disease *Provide treatment options by pulling information from the database and attaching links to online sources.

How we built it

We used Deep Learning pipeline to create such an incredible project, Steps are as follows : Adapt a pretrained model to our image classification problem AlexNet (Krizhevsky et al., 2012) + Transfer Learning Reshape the last layer: FC7 (4096 neurons) -> FC8 (neurons) Train the model with train-test ratio = (0.1 0.9) We used Pytorch to develop the model. and rest is in PPT

We used Figma as a foundation for the UI/UX interface

Challenges we ran into

we were trying to train model on local machine which was taking a lot of time then we side by side used Google Colab's GPU's !

Accomplishments that we're proud of

we were able to complete it in time

What we learned

working on Deep Learning Models! We also learned about the importance of treating crops and identifying the best and safest treatments as soon as a

What's next for CropStetho: creating combine our platform with drones used in agriculture to monitor various aspects of crops

With

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- svm

Try it out

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Submitted to



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Created by

- I worked on the ppt presentation by adding information and graphics. Also, I contributed to the understanding of the business perspective of our solution.



[Lannah S.](#)



AI & Vibe Coding enthusiast

- I worked on Deep Learning Model, Data Cleaning and The PPT.



[Amit Patel](#)





[Ziva Wernick](#)

Computer Science | Software development | Continuously learning

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Updates



[Ziva Wernick](#) started this project — [6 years ago](#)

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